Lab 2 – Smart City Product Specification

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Table of Contents

1 INTRODUCTION 3

1.1 PURPOSE 3

1.2 SCOPE 4

1.3 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS 5

1.4 REFERENCES 7

1.5 OVERVIEW 9

2 OVERALL DESCRIPTION 9

2.1 PRODUCT PERSPECTIVE 9

2.2 PRODUCT FUNCTIONS 10

2.3 USER CHARACTERISTICS 12

2.4 CONSTRAINTS 12

2.5 ASSUMPTIONS & DEPENDENCIES 13

List of Figures

Figure 1 - Major Function Component Diagram 10

List of Tables

Table 1 - Smart City Feature Description and Prototype Implementation…….. 11

1 Introduction

Cities are getting bigger and more crowded every year. The United Nations projects that by 2025, approximately 6.5 billion people will be living in urban areas (Luoma, 2016). This represents nearly 68% of the world's population concentrated in cities. Because of this rapid urbanization, people deal with longer commutes, increased traffic congestion, higher pollution levels and reduced quality of life. The way most cities are currently designed cannot handle this kind of growth, so problems like traffic jams, overcrowding at popular destinations, and poor resource management continue to worsen (Joyce, 2025).

Smart City is an application designed to address urban overcrowding by connecting residents to real-time community information and intelligent navigation tools. This application provides users and city officials with the start of a tool that will be used to improve the efficiency and accessibility to local information, thereby allowing both residents and planners to make better use of available data. The platform offers connectivity, as Smart City creates a network where residents can share and access hyperlocal updates. The application ensures community engagement, so residents can actively contribute to improving their urban environment through shared observations. Finally, it delivers clarity, because both residents and city officials will have access to timely, accurate information about local conditions and infrastructure needs.

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for the Smart City mobile application prototype. It is intended for the development

team responsible for implementing the system and serves as the primary technical reference for design and implementation decisions throughout the prototype development cycle.

This document covers the resident-facing features of the Smart City prototype, including account management, location-based community forum functionality, basic navigation via the Mapbox API, and the underlying data anonymity infrastructure. The Data Platform component, including analytics, data visualization, and long-term data infrastructure, is outside the scope of this document and will be addressed in a future specification.

1.2 Scope

The Smart City prototype will assist urban residents in navigating their local areas and communicating with their community. To demonstrate the core value of the real-world product, the prototype focuses on the resident-facing features that are most essential to the beginning of this goal. The Data Platform component has been deferred in its entirety due to time and scope constraints. Only the underlying database for residents and the anonymity of their data will be implemented in the prototype. This includes the users, sessions, forum posts, location data and notifications.

The prototype will allow residents to search for destinations and receive turn-by-turn navigation though the Mapbox API integration. Users will be able to access and post to the community forum, with posts tied to their current location and visible to other users within a 15-mile radius. The prototype will include account creation, login and basic account management. Alerts and notifications for local events and overcrowding conditions will be partially implemented using simulated data until the full notification pipeline can be established. Data anonymity protections will be full implemented in the backend to ensure the privacy of new users.

1.3 Definitions, Acronyms, and Abbreviations

Android: an open-source operating system used on mobile devices such as smartphones and tablets.

Application Programming Interface (API): a set of rules for two different programs to communicate with each other.

Application Layer: a layer where an application can effectively communicate with other applications on different computer systems and networks.

Backend: a computer system or application that is not directly accessed by users but is responsible for storing and manipulating data.

Bun: a fast JavaScript runtime and package manager used as the server-side execution environment for the Smart City backend.

Community Forum: a location-based platform within the application where residents share updates, report local issues, and engage in neighborhood discussions.

Crowdsourced Data: information gathered from a large group of people through digital platforms and user contributions.

CSS (Cascading Style Sheets): a stylesheet language used to define the visual design of user interfaces.

Data Layer: a layer of the application that contains all data generated and stored by users engaging with the system.

Docker: a platform designed to help developers build, share, and run applications in containers.

ElysiaJS: a TypeScript-first backend framework used to implement the application layer of the Smart City prototype.

Expo: an open-source framework and toolchain for developing and deploying React Native applications on iOS and Android.

Frontend: the portion of the application exposed to users for direct interaction.

GitHub: a hosting service for version control and collaborative software development.

iOS: a closed-source mobile operating system used in iPhones and iPads.

JavaScript: a high-level programming language used for frontend development.

Mapbox API: a third-party mapping and navigation service used to provide location services, routing, and real-time map visualization within the Smart City prototype.

MySQL: a relational database that uses SQL as its query language; used to store user, forum, session, location, and notification data.

Privacy-First Design: building systems such that, by design and by default, only the personal data necessary for a purpose are processed.

Presentation Layer: a layer responsible for rendering the user interface and facilitating user interaction.

React Native: a framework that allows for the building of cross-platform native mobile applications using JavaScript and TypeScript.

Structured Query Language (SQL): a widely used database language for inserting, querying, and retrieving data from relational databases.

TypeScript: a strongly typed superset of JavaScript used for backend and frontend development in the Smart City prototype.

Virtual Machine (VM): a digital version of a physical computer; used by Old Dominion University to host the Smart City database.

Visual Studio Code (VS Code): a source-code editor used as the primary development environment

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1.5 Overview

This product specification provides the hardware and software configuration, interfaces, and features and capabilities of the Smart City prototype. The following sections will provide detailed descriptions of each feature and their requirements for the prototype's implementation.

2 Overall Description

Smart City is a cross-platform mobile application designed to provide citizens with real-time urban information while enabling city planners and officials to collect valuable data for informed decision-making. The application addresses the growing challenges of urban congestion and overcrowding by combining navigation assistance, crowd monitoring, and intelligent route optimization with a location-based community forum into a unified platform.

2.1 Product Perspective

Third party software will be utilized to help provide some functionalities for the prototype, such as Mapbox which will contribute to the navigation and map search functionalities of the program. Old Dominion University's virtual machine via the Coder Workspace environment, will host the database using Docker containers. The architecture of the Smart City prototype will have a three-tier design, which include the presentation layer, an application layer, and a data layer. This design is shown in Figure 1.

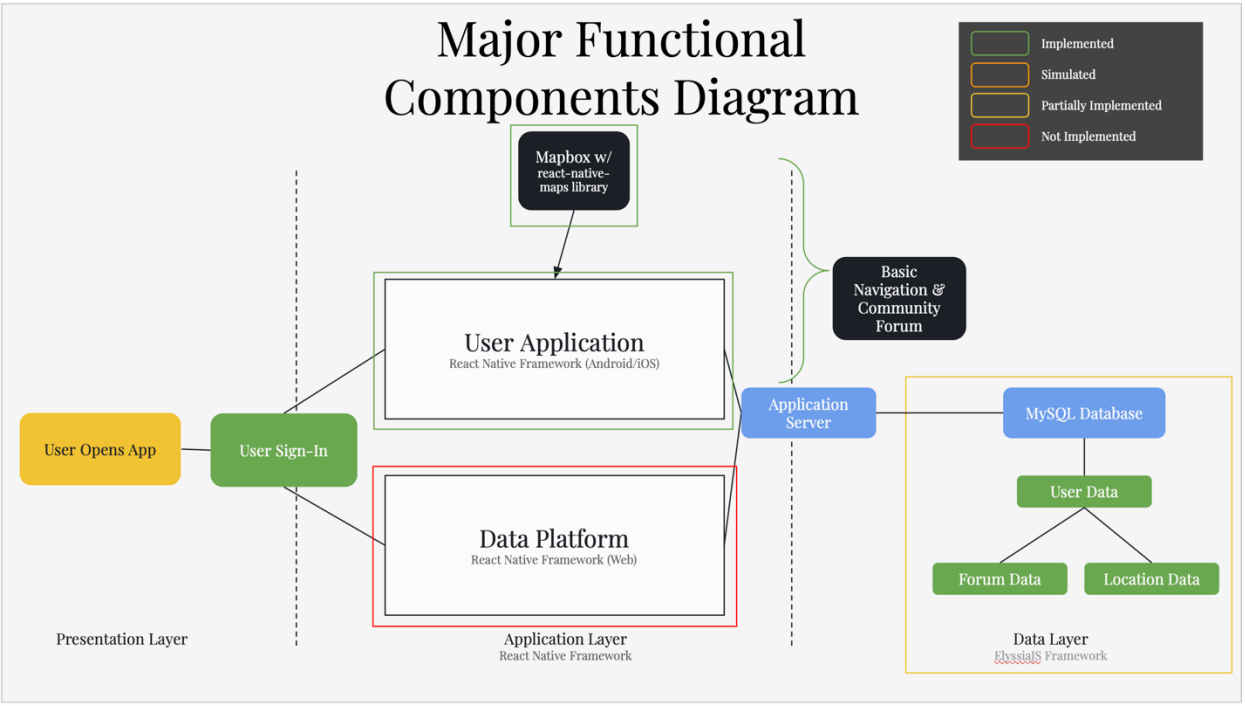

Figure 1 - Major Function Component Diagram

The presentation layer is where frontend development takes place using React Native to implement the user interface for residents. The languages used to build these interfaces are JavaScript, TypeScript, and CSS. The application layer features backend development using the ElysiaJS framework to implement the business logic and algorithms for each feature. The data layer is where database development occurs, using MySQL to handle data pertaining to users, user locations, forum posts, sessions, and notifications.

2.2 Product Functions

Most of the features of the real-world product will be fully implemented in the prototype, as can be seen in Table 1. For residents these features are Login/Authentication, Account Creation/Deletion, Manage Account Access/Permissions, Access Community Forum, Access to the basic navigation functionality, Alerts and Notifications, and Data Anonymity.

Category	Feature	RWP	Prototype	Reasoning
Account Management	Login / Authentication	Fully Implemented	**Fully Implemented**	Essential for prototype.
	Account Creation / Deletion	Fully Implemented	**Fully Implemented**	Essential for prototype.
	Manage Account Access / Permissions	Fully Implemented	**Fully Implemented**	Essential for prototype.
User Application	Access Community Forum	Fully Implemented	**Fully Implemented**	Essential for prototype.
	Navigation (Basic Routing)	Fully Implemented	**Partially Implemented**	The Dynamic Queue Navigation System was removed due to time and complexity constraints. Basic Mapbox navigation is retained.
	Alerts and Notifications	Fully Implemented	**Partially Implemented**	Feature is simulated until the full notification API pipeline is established.
	Parking Locations	Fully Implemented	**Not Implemented**	Eliminated due to scope constraints. May be reconsidered in a future version.
Data Platform	Central Information Hub (Database)	Fully Implemented	**Partially Implemented**	Database is implemented for storing user, forum, session, and location data. Long-term data platform infrastructure is deferred.

Category	Feature	RWP	Prototype	Reasoning
	Data Analytics and Insights	Fully Implemented	**Not Implemented**	Eliminated because it depends on the long-term data platform, which is outside the scope of the prototype.
	Data Visualization	Fully Implemented	**Not Implemented**	Eliminated for the same reason as Data Analytics and Insights.
	Data Anonymity	Fully Implemented	**Fully Implemented**	Essential for prototype; required for user privacy compliance.

Table 1 - Smart City Feature Description and Prototype Implementation

The features that will be partially implemented are the Dynamic Queue System and Alerts/Notifications. The navigation feature will be retains the basic routing through Mapbox but will exclude the dynamic queue optimization that was planned for the RWP as the complexity of that system exceed the available development time. Alerts/Notifications will be simulated until the full notification pipeline can be implemented. The Data Platform features have been eliminated from the prototype entirely, as they depended on a long-term data storage infrastructure that is outside the scope of this iteration. The Parking Locations feature has also been eliminated due to time and scope constraints.

2.3 User Characteristics

N/A

2.4 Constraints

N/A

2.5 Assumptions & Dependencies

N/A